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CSC 103 — Past Questions & Materials

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COURSE DETAILS**Module 1: Introduction to Computer Systems****Unit 1: Fundamentals of Computer Systems****CONTENTS**

- 1.0 Introduction
- 2.0 Intended Learning Outcomes
- 3.0 Main Content
 - 3.1 Definition of a Computer
 - 3.2 Descriptions of Computer Systems
 - 3.3 Classifications and types of Computer Systems
 - 3.4 Historical Generation Advancement of computer systems
 - 3.5 Comparison between Hardware and Software
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**1.0 Introduction**

Today, almost everyone uses computers directly or indirectly to solve diverse problems ranging from word processing to internet-based applications. For instance, your access to this course material is through the use of computers and computer applications. This course is about the study computer hardware, which is the physical part of the computer system. This module addresses the introductory theories about computer systems. It also deals with interconnectivity and relationship with the major modules of a standard computer system. In this Unit, we shall discuss some fundamental concepts relating to a computer system. The Unit also explains the computer device and system configuration.

**2.0 Intended Learning Outcomes**

By the end of this Unit, you will be able to:

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- Understand the concepts of computer systems
- Observe the interconnectivity of subsystems in a computer system



3.0 Main contents

3.1 Definition of a Computer

A computer can be described as an electronic device that can receive data, process the data, and produce the result as the outcome. Conventionally, the received data is known as input while the result of the processing is known as output or information. A computer also has the capability of storing data and/or information in its memory for future use. This relationship is illustrated in Figure 1.

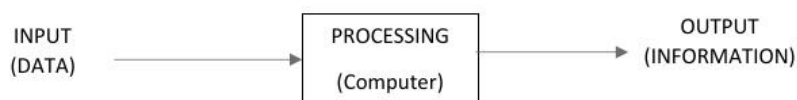


Figure 1 Computer Device Data Flow Process

Basically, a computer is divided into two major parts, namely, hardware and software. This configuration is like the division of human being into body and spirit/soul. That is, the body is the hardware while the software is the spirit or soul. While the hardware, the physical or most visible part is tangible, the software is intangible. Normally, the software resides and operates within the hardware. Essentially, hardware is the physical part of the computer while the software is the code that runs on the computer.

Computers are capable of processing numerical data in form of 0's and 1's or simply in binary digital format. This digital format is the basis of the machine language which differs from human-readable languages. They can solve repetitive computational jobs with high level of accuracy and timeliness depending on the software capability.

3.2 Description of a Computer System

A computer system involves a complete suite of hardware and software resources as well as the users that operate the computer system. While a computer system does the basic function of a computer device, different kinds of computer systems can handle certain specific additional functionalities. This will be addressed later in this Module.

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Ideally, a full-fledged computer system is made up of users, application software, systems software, operating system and hardware as shown in Figure 2.

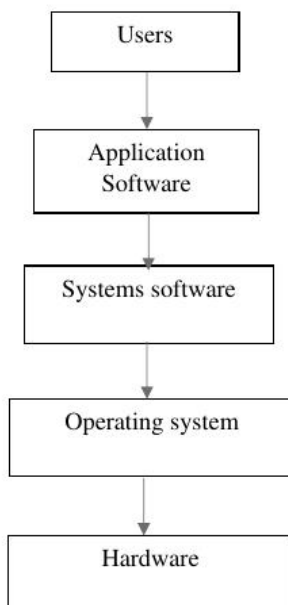


Figure 2 Architecture of a Computer System

In Figure 2, each of the computer system parts is known as a subsystem. In this case, a subsystem is made up of more than one component. For instance, the users' subsystem comprises expert users, programmers/developers, and end-users.

The main focus of this course is the hardware subsystem. It is the bedrock of other subsystems.

3.3 Classifications and Types of computers

You must have been aware that many kinds and types of computers exist for different applications and computational needs. Can you identify some of these? Essentially, computers while maintaining their basic functionalities, can be categorized according to their processing power, internal design, applicability, and physical size. These categories are as follows:

➤ **Personal computers**

Personal computers, popularly shortened as PC, are common, cheaper, versatile, small and portable in sizes. They are single-user computer based on different kinds of microprocessors. Laptops are kinds of PCs that consume less power due to their miniaturized integrated internal components. We shall discuss about processors and their types later in this course.

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Today, many students, office workers and business people use personal computer to carry out many tasks. Such tasks include word processing, graphical designs, desktop publishing, accounting operations, spreadsheets, video editing, teleconferencing among others.

It should be noted that the specifications of personal computers have evolved over the years due to explosion in human population, advancement in software and hardware components.

➤ Workstations

These computers are relatively more powerful than PCs in terms of computational capability. They are also single-user based. Although, a workstation is like a personal computer, however it has a higher specifications and better quality. These high-end computers are often used to execute complex and large computational jobs in lesser time, so they are usually more costly than PCs. These types of computers are found in research laboratories, research institutions, universities, Space, automotive, and manufacturing industries.

Note that the specifications of workstation computers have also evolved over the years due to software and tasks requirements.

➤ Minicomputers

Minicomputers are multi-users-based computers. They support a large array of users concurrently and are usually able to handle larger complex or/and large computational jobs. PCs and workstations are usually connected to minicomputers in a network for the joint execution of tasks, processes, and programs. Minicomputers are not portable, they belong to large organizations such as the Nigerian National Space Research and Development Agency (NASRDA), National Identity Management Commission (NIMC), Central Bank of Nigeria (CBN), national astronautics and space administration (NASA), European Space Agency (ESA), among others. During your next industrial training, you may wish to visit such organization to see such computers.

It should be noted that the specifications of minicomputers computers have also evolved over the years due to software and hardware advancement.

➤ Mainframe

Mainframe computers are larger, expensive and more powerful multi-user systems. They simultaneously support thousands of users than minicomputers. They have higher specifications in terms of processing speed and storage. They are quite huge and expensive. Mainframe computers are mostly found in large corporations,

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enterprises and government organizations where many IT professionals and workers use them simultaneously to solve grand challenge computational problems

They are designed to perform large numbers of calculations for governments and large enterprises

It should be noted that the specifications of mainframe computers have also evolved over the years. International Business Machine (IBM) is one of the earliest manufacturers of computer systems. Figure 3 shows the System z9 computer which was a line of mainframe manufactured by IBM in 2005.



Figure 3 IBM System z9

➤ Supercomputer

Although, these computers are similar to mainframe, they are extremely the fastest and can be used to perform hundreds of millions of instructions per second. They are designed to solve grand challenge problems that are often data- and computationally intensive, such as DNA sequencing/human genome, Space exploration, Mars mission, national population census, national electoral system, national database systems, banking applications and database, health informatics, etc. A typical supercomputer can solve up to ten trillion individual calculations per second.

It is important to note that the specifications of supercomputers have also evolved over the years. Usually, supercomputers are the most powerful computers that exist at a time in human history. Today, China and USA are the leading manufacturers and users of supercomputers in the world. This fact is available at the <https://www.top500.org/> website.

Till early 2021, Fugaku, the Chinese exascale systems was known to be the world's fastest supercomputer on Top500. As at this press time, the Intel and Cray of the Argonne National Laboratory is working hard on the design and development of

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Aurora as the fastest supercomputer to be commissioned in late 2022. The project is under the sponsorship of the United States Department of Energy. Figure 4 show a typical structure of a computer system.



Figure 4 Computer system

These categories of computers spread across the generational phases and development of computer systems of all history.

3.4 Historical Generations Advancements of Computers

Computer history specifically dated back to 1833 when Charles Babbage, a mathematician, invented the first runner of digital programmable and general-purpose computers. Barely, a century later, Electronic Numerical Integrator and Computer (ENIAC) was invented by John W. Mauchly and J. Presper Eckert, this computer was the first general-purpose electronic computer.

Obviously, the entire history of mainly computer hardware advancements were distinctly partitioned into time-frame, periods, or generations according to a specific range of advancements. The generations are as follows:

a) First Generation: 1940-1956

This generation showcased the use of vacuum tubes for logic circuitry through the flow of electrical current. Magnetic drums were used for data transfer and storage. During this period, machine language was developed for systematic programming. ENIAC was among the systems developed during this period. The drawbacks of the

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systems include slower speed, excessive heat, and high cost. Figure 5 shows a typical first generation computer.



Figure 5 First Generation Computers

b) Second Generation: 1957-1963

In this generation, transistors were used as the major switching devices thus making computers smaller, faster and energy-efficient. The programming format was assembly languages. Figure 6 shows a typical second generation computer.



Figure 6 Second Generation Computer

c) Third Generation 1964-1971

Integrated circuits (IC) were developed and used as the switching devices for the flow of data. A single IC contains several numbers of transistors, capacitors, and resistors in the entire circuitry. Thus, computers became smaller in size, faster in

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processing and generally efficient. High level programming languages were also introduced during this period. Figure 7 shows a typical third generation computer.

Prominent computers in the third generation were PDP (Personal Data Processor), IBM-168/360/370 series, Honeywell-6000 series



Figure 7 Third Generation computer

d) Fourth Generation Computers 1972-1990

The fourth-generation computers were based on the successful invention and development of microprocessors as the central processing units (CPU). Personal and portable computers were manufactured due to the production of miniaturized computing components. As a result, the Very Large Scale Integrated (VLSI) circuits technology was implemented in the design of the microprocessors and other kinds of switching devices. Intel was the first company to design and develop microprocessors. The details of microprocessor or CPU will be discussed later in this course. Efficient high-level programming languages were also developed such as Java, C++, C#, VB.NET among others. Figure 8 shows a typical example of the fourth Generation Computer.



Figure 8 Fourth Generation Computers

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e) Fifth Generation Computers 1990 to date

The computers of this generation are designed based on superconductors, VLSI, miniaturized components, multi-core technologies, and artificial intelligence. Future computers will advance more on machine learning, expert systems, natural language processing, neural networks, advanced parallel computation, robotics, and fuzzy logic designs. Virtual reality, fault-tolerance, and artificial intelligence will dominate the hardware and software designs of fifth-generation computers.

The robots are being manufactured as computing devices that can replace human beings to work in difficult, dangerous, and boring environments with less human intervention. Examples of these robots are satellites, Sophia, robocop, mars rover, among others. Find out what these robots do and identify other kinds of robots. Figures 9 and 10 show examples of the fifth generation computers.



Figure 9 Fifth generation computer - Laptop



Figure 10 Fifth generation computer - Robot

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3.5 Comparison between Hardware and Software

Table 1 in this section shows the comparison between hardware and software.

Table 1 The comparison of hardware to software

Hardware	Software
Hardware is a tangible physical parts computer system	Software is a set of instruction or codes that control the functions of a computer
Hardware devices are manufactured	Software are developed
Hardware cannot function without software.	software needs hardware to operate
As Hardware are physical electronic devices, we can see and touch hardware.	We can see and also use the software but can't actually touch them.
Von Neumman architecture for computer framework	This adopts software models
computer viruses do not affect computer	Computer viruses can corrupt codes
Yes, hardware can do not bare metals	It can be downloaded, copy and transfer
If hardware is damaged, it can be replaced with a new one	When software is damaged, corrupt it can recovery from backup or the nendows
Examples include Keyboard, Mouse, Monitor, Printer, CPU, Hard disk, RAM, ROM etc.	Examples include Ms Word, Excel, Power Point, Photoshop, MySQL etc.

4.0 Conclusion

A computer has been described as an electronic device that is capable of receiving data, processing the data and generating the output result in a timely manner. A computer system comprises software and hardware. Some basic

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categories of computers include supercomputers, mainframe, minicomputer, desktop computers, and personal computers.

5.0 Summary

In this Unit, you have learned about some fundamental concepts of a computer system.

6.0 Self-Assessment Exercises

- What is a computer?
- Mention some tasks which computers can be used for
- Describe a computer system
- Advancements in computer hardware are the major causes of different generations in computer history. Discuss.

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<https://www.top500.org/>

Unit 2: Computer Software

Content



1.0 Introduction

Do you realize that computer hardware cannot work without software? You have learned from the Unit 1 that software is the soul or intangible component of the computer system. When you drive your motor car, you are able to control the operations and direction of movement of the car. That is what a software does exactly to the computer hardware. In this Unit 2, we shall study about the detailed professional description of software and the types that are available to the users. It is certain that you have used some of several kinds of software directly or indirectly over the years. Relax well as we discuss this important subject matter in computer systems.

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2.0 Intended Learning Outcomes

By the end of this Unit, you will be able to:

- Understand the definition and relevance of computer software
- Know some types of computer software



3.0 Main Content

3.1 Descriptions of Computer Software

Software can simply be described as a set of codes or programs involving a combination of well-defined instructions, scripts, procedures, documentations, and data that can manipulate and execute predefined tasks on a computer system. These tasks also include the control of functionalities of the computers or computing devices in handling jobs according to the programming. The computing devices include PCs, tablets, PDA, Automated teller machines, mobile phones, smart TVs, and other microcontroller-based embedded systems or smart devices.

Basically, computer hardware devices cannot operate or function without software. This inter-relationship between hardware and software is as shown in figure 2.0. You need to also understand that a software suite comprises programs, also known as a set of codes, design specifications, documentation, and users' manuals.

3.2 Types of Computer Software

Since the inception of the computer age, there have been several kinds of software serving different purposes ranging from hardware control, users' operations, management of hardware to the entire system control. Most times, the users, developers, manufacturers and programmers determine the availability of different kinds of software that are used in the computer systems.

Generally, software can be classified into application software, system software utility software, operating system and firmware. All these forms of software are available in computer systems and software engineering environment so as to support ICT facilities, computer users, programmers, developers among others. Figure 11 shows the classifications of software. Let us go through all these now.

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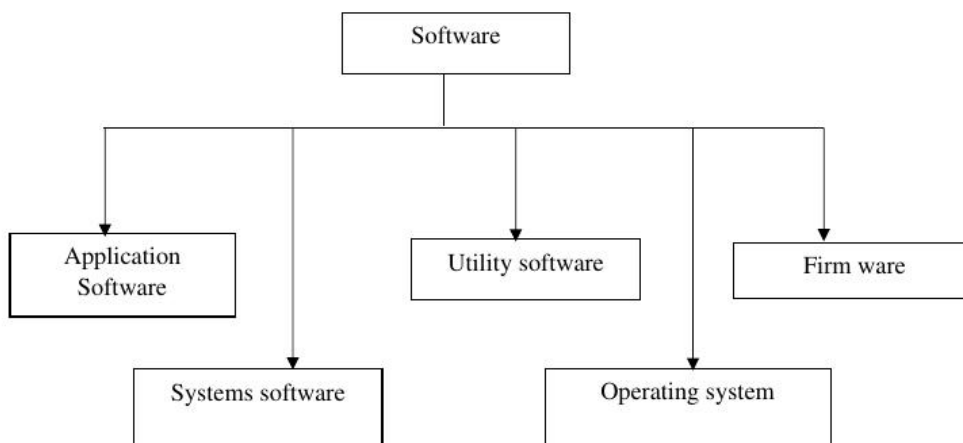


Figure 11 Classifications of Computer Software

Operating system

An operating system software is a software that interfaces between system hardware, application software and the users. It is analogous to a company secretary or receptionist that interact between the guests and other workers or offices of the organization. On arrival, the guests may be directed to appropriate employee or office that can meet the needs of the guests. Thus, without the roles of operating systems the computer hardware cannot be operated, controlled and manipulated by the users. The effective communication between hardware and users is very important for both parties.

Essentially, the operating systems enable the management of all the computer resources. These resources include memory, files, processes, peripheral devices, security features, processors, storage devices, networking functionalities, job scheduling among others.

Operating systems (OS) can be broadly categorized as single-users and multi-users. While a single-users OS allows one user at a time, a multi-users OS allows more than one user to operate a computer at a time. We also have single-tasking and multi-tasking operating systems. Tasks are processes or jobs running by the computer central processing unit. Now, you should be able differentiate between single-tasking, and multi-tasking. A single-user OS may be either single-tasking e.g. (MS Windows 95) or multi-tasking (e.g. MS Windows 7, 8, 9,10,11). MS here means Microsoft, one of the prominent software developer companies in the world today.

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Examples of single-users operating systems are MS-DOS, Windows 95, Windows NT, Windows 2000, Microsoft Windows 7, 8, 9, 10, 11, etc.

Examples of multi-users operating systems are Mainframes, IBM AS400, Linux & Unix Distributed OS, etc.

Application software

An application (app) software is a computer program suite specifically designed and developed to solve a certain problem, perform a job or execute an activity in various human endeavours. Mostly, applications manipulate or work on data, text, numbers, audio, graphics, and video with respect to the objectives of the operations.

Examples of application software are MS Word, Excel, PowerPoint, Google Chrome, Photoshop, MySQL etc.

Firmware

Firmware is a kind of software that enables the low-level control for a specific micro-controller-based hardware device. Firmware software are mostly used for embedded systems operations and control. Today, many electrical/electronic appliances such as washing machines, air conditioning units among others have firmware running in their computing systems.

4.0 Conclusion

It is important to note that a computer system or computing device is broadly divided into hardware and software. While hardware refers to the physical components or parts, the software is the soft part, non-tangible, or simply codes and programs that drive the hardware. The computer hardware machines cannot function without the software. Thus, the software helps to achieve the functions a computer. Types of software include systems software, operating systems, application software and utility software.

5.0 Summary

In this Unit, you have learned about software, types of software and software applications.

6.0 Self-Assessment Exercises

- ✓ Give some analogies of computer software
- ✓ What are the roles of computer software?
- ✓ State some examples of computer software
- ✓ Distinguish between firmware and operating systems

7.0 References and Further Reading

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Gagne, Silberschatz Galvin (2012). Operating Systems Concepts. New York: Wiley. p. 716. ISBN 978-1118063330.

<https://www.webopedia.com/definitions/software/>

Module 2: Computer Hardware Architecture

Unit 1: Von Neumann Computer Architecture

Content

1.0 Introduction

In the last Module, you learned about computers functions and the types of computers. In this module, we shall study deeper the architectural design, layout, nature, and structure of a computer system. In 1945, the well-known and most acceptable standard architecture for all digital computer systems was designed and published by John Von Neumann, a Hungarian mathematician who also doubled as a physicist. This man was thus the architecture is tagged Von Neumann architecture.



2.0 Intended Learning Outcomes

By the end of this Unit, you will be able to:

- Understand the marketecture of computer systems
- Peruse the detailed component of the computer system



3.0 Main Content

3.1 Von Neumann Computer Architecture

Von Neumann architecture described the digital-based computer hardware as a system comprising the following major parts

- System Unit
- System board
- Central Processing unit
- Ports
- Input Devices
- Output Devices
- Memory

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- Secondary storage
- Communication devices

The functionality of these aforementioned parts is a similitude to the joint operation of the human body. Our brain is the central processing unit, our legs and hands are the input/output devices. Can you guess what the memory devices may refer to?

Figure 12 shows the pictorial description of the Von Neumann architecture that serves as the basis for all digital computer systems.

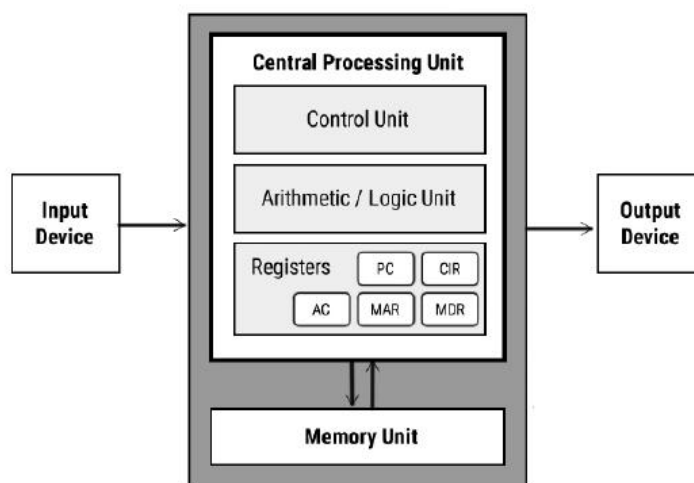


Figure 12 Von Neumann Computer Architecture

In Figure 12, the central processing unit, popularly known as CPU for short, comprises of the control unit, arithmetic/logic unit and processor registers. The computer system operations are also controlled by the peripheral devices which are basically known as the input and output devices.

We shall discuss the details of these component parts in the following modules

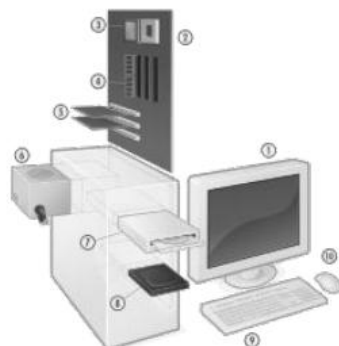
3.2 System Unit

The system unit houses all the internal components of a computer system. These internal components include the system board, also known as the motherboard, the power supply

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unit, fixed hard drives, communication buses among others. Figures 12 and 14 show the illustrations of computer system unit.



Label

1. Monitor
2. Motherboard
3. CPU
4. RAM
5. Expansion cards
6. Power supply
7. Optical disc drive
8. Hard disk drive
9. Keyboard
10. Mouse.

Figure 13 Personal computer hardware



Figure 14 System unit of a personal computer

The entire system unit of a personal computer is often located in a plastic or metallic case or enclosure. Such personal computers are desktops, laptops, and palmtops. The desktop PC case, which is usually designed to be placed on desks, can either be in the tower or flat shape. Nowadays, there are compact, all-in-one desktop system units having all the peripheral devices integrated together. Such all-in-one systems are manufactured by Dell, HP, and Apple (iMac). Most high-performance computer servers and workstations are available in bigger, rack and tower shapes.

Power supply unit

The computer power supply module which is located in the system unit case enables the conversion from 100-240V alternating current to low-voltage direct current voltage to power the internal components according to the requirement specifications. Some

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systems like laptops have the in-built batteries for power backup when utility power supply fails.

System upgrade

The computer system upgrade often refers to improvement on the hardware and software resource specifications so as to enhance the system capability and overall performance. For instance, in terms of hardware, the RAM size, HDD/SSD capacity, visual display unit, and CPU can be enhanced. In most cases the software and hardware advancements must match. For instance, it is advisable that the 64-bit software should run on the 64-bit processor, otherwise there may be a speed mismatch, error in computation, or operation failure.

An expansion card, which is a printed circuit board, can be inserted plugged via its socket into an expansion slot of a computer motherboard through the data bus according to the required need of the computer system. Graphics, visual display unit, sound, networks subsystems all have either in-built connections on the motherboard or expansion cards.

3.3 Computer Communication Buses

The name bus is an analogy derived from vehicle buses to transport human beings from one place to the other as shown in Figure 15. Computer bus lines, wire, or cables are electrical data paths through which bits of information are transmitted between the CPU and other components. They are communication cables and devices that enable data transfer among the internal and external components or parts of a computer system. Buses are also both hardware and software-based. Hardware buses include wires, optical fibre, etc. while software-based buses include communication protocols. The CPU word size, also known as the bus size refers to the number of bits that can be transmitted at once. In general, this should be the same as. Figures 16 and 17 show some common types of communication buses within computer systems.

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Figure 15 A vehicle bus for human transportation



Figure 16 A computer bus for data transfer

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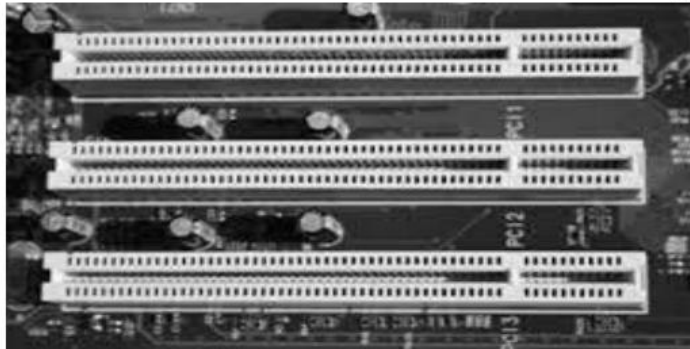


Figure 17 Computer bus for data transfer between the motherboard and RAM

Historically, computer buses were initially made up of parallel electrical wires involving several connections of hardware devices. Today, computer communication buses also include parallel and serial physical arrangements that support the logical functions. Computer buses are simply wired in a daisy or multidrop arrangement as in universal serial bus or switch connections.

4.0 Conclusion

The Von Neumann computer architecture is the most acceptable standard of computer systems parts and connections upon which computer digital systems are based. The CPU, peripheral and storage devices are the main parts well revealed by this standard architecture. The computer communication buses are essential for data transfer among the computer hardware resources.

5.0 Summary

In this Unit, you have learned about computer hardware architecture. Lessons learned include the Von Neumann architecture with respect to the interconnectivity of the major parts of computers via different kinds of data buses.

6.0 Self-Assessment Exercises

With the aid of a well-annotated diagram, explain the main components of the Von Neumann computer architecture

Explain the concept of data communication buses

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Unit 2: Central Processing Unit

Content



1.0 Introduction

In this Unit, we shall explain the roles and functions of the main internal component of the computer system, which is the Central Processing Unit (CPU). The CPU is metaphorically referred to as the hearth-beat or brains of the computers. The CPU carries out the execution of computer instructions or programs and the manipulation of data. It also controls the functions of other parts of the computer so as to enable the smooth running of the entire computer system. The CPU is also known as the processor, microprocessor or central processor of the computer.



2.0 Intended Learning Outcomes

By the end of this Unit, you will be able to:

- Comprehend the central processing unit of the computer systems
- Understand the roles and functions of system CPU
- Know some types of CPUs



3.0 Main Content

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3.1 Central Processing Unit

The central processing unit (CPU), also known as the processor is an electronic chip of an integrated circuit that control the computer and also performs arithmetic logical operations. The main function of the CPU in a computer system is to receive, implement and execute all the instruction codes received from the computer hardware and software. The CPU on the satellite onboard computer is popularly known as the central terminal unit. The CPU on the graphic adapter of an image processing computer system is known as the graphic processing unit. All these variants have the same functions and architecture. Figure 18 shows the computer CPU.



Figure 18 The computer CPU

The CPU comprises of the control unit (CU), arithmetic and logical unit (ALU), processor registers, and the program counter. Figure 19 shows the block diagram of the CPU and its internal components.

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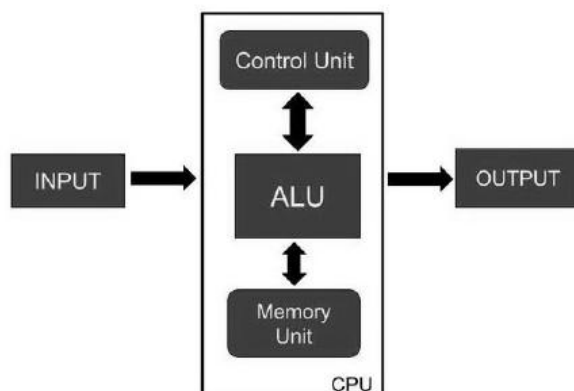


Figure 19 Block diagram of the CPU and its internal components.

How the CPU works

The figure shows that the CPU receives data through input devices and stores it in the memory. The control unit thereafter extracts the data from the memory according to the instructions. Consequently, the ALU executes necessary arithmetic and logical operations on the data manipulations. These operations include addition, subtraction, multiplication, and division. The result of the computations is stored in memory and finally sent out as output via the output devices. The control unit basically deciphers and carries out the pre-defined instructions or programs.

The word size refers to the number of bits of data a CPU can process at once. Today, due to advancement in hardware technology 64 bits is the standard word size for CPU's used in personal computers, the older CPU used 32 bits. The overall speed of a CPU in executing programs is proportional to its word size.

Different CPUs have different types of instructions, so software made for one type of CPU will not run on other kinds.

The Intel Pentium and PowerPC are common examples of processors for IBM-PCs and Macintoshes respectively.

Generally, the CPU is usually located at the center of the computer motherboard as shown in Figure 20. The CPU is usually cooled by a heat sink and fan often mounted on top of it. Most newer CPUs include an on-die Graphics Processing Unit (GPU) and co-processor for graphic and mathematical processing respectively.

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Figure 20 The CPU inside the computer motherboard

3.2 ALU

The arithmetic logic unit (ALU) component uses a combinational digital circuit to perform bitwise operations and arithmetic calculations on integer binary numbers. The ALU does not operate on other fixed- or floating-point numbers. It is a key component structure of several kinds of computing circuits, including the CPU and GPUs. The ALU operates on the operands and codes which are the inputs as the data for the system. Subsequently, this generates the output of the executed operations through the status input/output which provides information about the operations. This is achieved by the in-built external status registers.

3.3 Micro-controller or system-on-chip

The microcontroller is a complete computer on one chip, simply known as system-on-a-chip (SoC). It is a small chip with all the required electronic components, CPU, RAM, storage, and other circuitry in a specific device. Examples of the applications include a smartphone or portable computer, on a single slab of an integrated circuit, car, microwave, thermostat among others. For instance, arduino computer on a board has a microcontroller chip.

4.0 Conclusion

The CPU is an integral part of the computer hardware. It is conspicuously located at the centre of the motherboard. The computer CPU enables the control of the hardware resources and also carries out the arithmetic and logical operations of the system. The micro-controller is simply the complete computer on a small single slab of an electronic chip.

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5.0 Summary

In this Unit, you have learnt about the CPU and micro-controller.

6.0 Self-Assessment Exercises

Describe the functionalities of the CPU

Identify the components of the CPU

Define a micro-controller

7.0 References and Further Reading

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Unit 3: Computer Data Storage and Memory Devices

Content



1.0 Introduction

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In this Unit, we shall deal with computer data memory or storage devices. These are significant media components of the computer system used for recording or retaining all forms of digital data. Every computer system requires several different levels of memory for storing system data, users' data, programs, and instruction sets either temporarily or permanently. The two keywords here are memory and storage, which slightly refer to different things. While the memory devices store data for a short-term for immediate access, the storage devices store data long-term for permanent access. Long-term storage is also known as persistent storage duration. The data, in this case, refers to documents, applications, programs, codes, and operating systems.

Computer memory can be classified into either primary and secondary or fixed and removable devices.

2.0 Intended Learning Outcomes

By the end of this Unit, you will be able to:

- Understand the computer memory organization and architecture
- Know some types of memory storage devices



3.0 Main Content

3.1 Primary Main Memory

The primary memory chips which are directly accessible by the CPU, is the main primary memory of the computer. The access speed of the main memory is faster than any other kinds storage devices. Examples of primary memory devices are Random Access Memory (RAM) and Read Only Memory (ROM). Both of them are made up of solid-state materials.

Figure 21 shows different types of memory and storage devices available in computer systems. Let us discuss some of these storage devices in detail.

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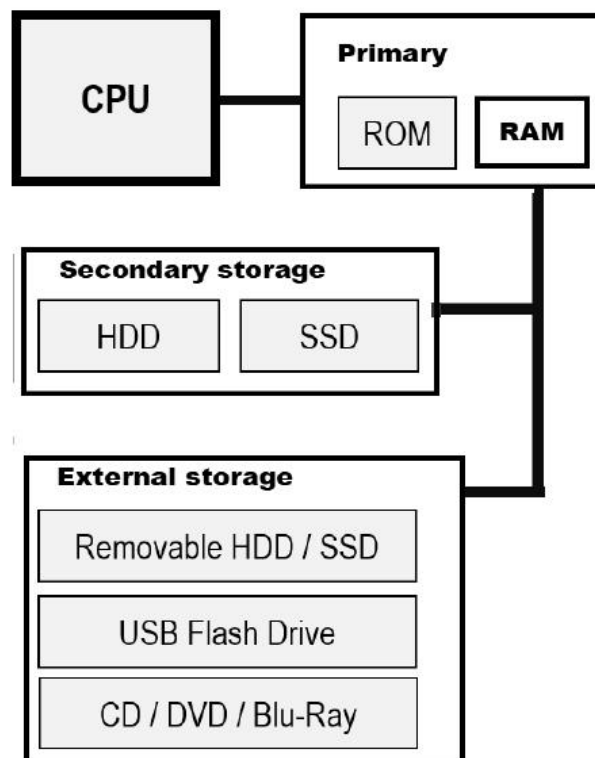


Figure 21 Computer memory and storage devices

Random Access Memory

Random Access Memory (RAM), is a main memory or primary storage that has capability of holding the loaded data, instructions, operating system and all running application, processes and other programs only during execution, that is, when the computer system is running and powered by electricity. RAM behaves like a scratchpad memory or whiteboard. temporary, it is a working storage area of bytes under the supervision of the CPU. RAM stores entire data temporarily. For instance, if you type your CV on MS Word without saving it with a file name, it remains inside the RAM while your computer keeps working. If the power is turned off in the process, your data and program will be erased automatically. Then you will lose your documents or data. This means RAM is not persistent, its contents are dependent on power supply. Therefore, always save up your document while working and making updates.

A good feature of RAM is about faster speed of access because its contents are being actively manipulated by the CPU. It is able to access and retrieve the value of any

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particular byte in a few nanoseconds (1 nanosecond is 1 billionth of a second). RAM is more than 1000x faster than the fastest secondary storage.

Usually, RAM is designed and manufacture with a few chips packaged together onto an integrated circuit, a popular example of this is the dual inline memory module (DIMM) that can be inserted into the motherboard socket as shown in Figure 22.

The two common types of RAM are static RAM (SRAM) and dynamic RAM (DRAM).

a) SRAM

This is a kind of RAM that uses flip-flops transistor or Mosfet (MOS) as switching devices. Its access speed is fast but more expensive. Example of SRAM is cache memory.

b) DRAM

Dynamic RAM uses capacitors as switching devices to store. The capacitors charges when data is 1 and do not charge when the data is 0. Unlike SRAM, DRAM requires refreshing circuits for the data refreshment so as to avoid current leakage. Usually, it is slower but cheaper than SRAM. Most computer main memory is made up of DRAM due to higher access time.



Figure 22 Random Access Memory in a motherboard

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Read-Only Memory

The Read-Only Memory (ROM) chip is a type of primary memory that stores the basic input/output system (BIOS) information that runs when the computer is booting or starting. They are non-volatile memory that generally contains instructions for booting the computer or loading the operating system. This information is in boot and power management firmware. Modern motherboards apply Unified Extensible Firmware Interface (UEFI) instead of BIOS.

The flash chips are non-volatile devices that do not require electricity. They are used in computers, cell phones, digital cameras, etc.

The cache memory is special high-speed memory that temporarily stores instructions and data that the CPU is using frequently, Thus speeding up the processing. The Level 2 or external caches generally range in size from 64 Kilobytes to 2 Megabytes.

3.2 Secondary Storage Devices

A secondary storage device can be described as a persistent or non-volatile medium that can save up data for a long time until it is overwritten, deleted, or corrupted. These storage devices can either be fixed or removable. Examples of secondary storage devices are Hard Disk Drive (HDD) and Solid State Drive (SSD). The removable USB flash drive and optical compact disks can be used to copy or transfer data between computers. The fixed drives are usually permanently connected to the system unit of the computer. HDD and SSD are often connected to the motherboard through communication cables. HDD is capable of using its read/write heads to store digital data on a magnetic surface of a rigid plate. HDD physical size is usually 3½ and 2½ inches for desktops and laptops respectively.

These types of storage devices shall be discussed as follows;

Compact Disc, DVD, Blu-ray are examples of optical external storage devices while USB Flash Drives are made up of solid-state materials. Flash drives as also known as thumb drives or USB-key.

Difference between Hard drives and Solid State Drives

A solid-state drive (SSD) is a secondary storage device that implements an array of integrated circuits to store data persistently. Hard disk drives are found in virtually all older computers, due to their high capacity and low cost, but solid-state drives are faster and more power-efficient but more expensive. Flash drives are removable and external SSD. They consume less power, and they are usually faster and cheaper than hard disk drives. Technologically, hard drives store bytes of data as a magnetic pattern on a spinning metal disk in high pitch spinning sound. The solid-state drives that store bytes on flash chips, do not have spinning parts but use silicon chips and electrons.

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Today, high-performance computer clusters use disk array controllers for greater reliability, security, and massive storage. Figure 23 shows Hard Disk Drive for persistent storage.



Figure 23 Hard Disk Drive for persistent storage

Figure 24 and 25 shows USB flash drive or Optical disc which can be used to transfer data between computers via the USB port. They are compatible with most hardware and software specifications.



Figure 24 Removable Flash Drive for persistent storage

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Figure 24 Removable Flash Drive for persistent storage

Flash drive

Figure 25 shows the SD Card which provides storage in devices like cameras. This looks similar to the USB flash drive.



Figure 25 SD memory flash

Other types of removable secondary storage devices are as follows

Compact disc (CD) is a circular thin plated glass and plastic polycarbonate material of a standard size of 12 cm with a hole in the center of about 1.5 cm and 1.2 mm in thickness. The CD has a storage capacity of 600 MB to 700 MB of data, this technological product is much older, the new ones will be discussed shortly. CD uses optical laser technology instead of magnetic technology to store its contents. Figure 26 shows bottom surface of a 12 cm compact disc.

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Figure 26 Bottom surface of a 12 cm compact disc

Types of Compact discs are:

- a) Compact disc Read-Only Memory (CD-ROM): The contents of the CD-ROM cannot be deleted by any means. Only the publisher is allowed to access the data imprinted on this CD. It is mostly used to store or copy small size documents or software applications.
- b) CD-Recordable (CD-R): The contents can be stored once and read several times. Just like CD-ROM, its contents cannot be deleted or overwritten.
- c) CD-RW (CD-Rewritable): The contents of CD-RW can be erased or rewritten several times.
- d) Digital Video/Versatile Disc (DVD): This type has a higher storage capacity ranging from 4.7GB to 17GB depending on whether it is a single or dual-layer format. The storage capacity of a DVD with a one-sided layer is 4.7 GB, the one-sided double layer is 8.5 GB, the double-sided layer is 9.4 GB, and the double-sided double-layer is 17 GB.
- e) DVD-ROM: The contents of this type of media cannot be written on or erased by the user. It is mostly used for distributing proprietary software or other kinds of applications.
- f) DVD-R / DVD+R: These two different types of discs, DVD-R (DVD minus R) and DVD+R (DVD plus R) are recordable once.
- g) DVD-RW / DVD+RW: The contents can be re-written several times.
- h) DVD-RAM: This is a rewritable disc that can alter its contents several times. It functions like hard disks.

A Floppy disk is an old, portable, and removable platter storage device that was made of magnetizable mylar plastic. The data is stored in concentric rings called tracks on either side of the platter. The last kind of floppy disk was a 3½ inches platter in a hard plastic case that holds 1.44 Megabytes of information. A Zip disk, on the other hand, could hold up to 250 Megabytes.

Magnetic tape, which was developed in Germany in 1928, is a medium for magnetic wire data recording. It was made of a thin, magnetizable coating on a long, narrow strip of plastic film. Tape recorders and video tape recorders are used to record and playback audio and video respectively. The tape drive stores computer data on magnetic tape.

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4.0 Conclusion

The storage or memory devices are integral parts of a computer system. The devices can be categorized into primary and secondary which can be either fixed or removable. RAM and ROM are primary memory while HDD and SSD are types of secondary non-volatile storage devices. The removable storage devices are USB drives, flash drives, and compact disc drives. The required types and capacity of drives needed for a computer system is a function of the well-configured system specification, cost, and the desired performance.

5.0 Summary

In this Unit, you have learned about different kinds of memory and storage devices. Quite some examples were also stated.

6.0 Self-Assessment Exercises

Distinguish between persistent and non-persistent storage of data

Differentiate between Hard disk drive and solid-state drive

List some types of compact discs you know

Why is the manufacturing of floppy drives discontinued?

7.0 References and Further Reading

Liebowitz, Kusek, Spies, Matt, Christopher, Rynardt (2014). VMware vSphere Performance: Designing CPU, Memory, Storage, and Networking for Performance-Intensive Workloads. Wiley. p. 68. ISBN 978-1-118-00819-5.

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Module 3: Computer Peripheral Devices**Unit 1: Input Devices**

Content



1.0 Introduction

In the previous modules of this course, you have learned about various hardware parts of computer systems as shown in the John Von Neumann architecture. Peripheral devices, which involve input and output devices are significant parts of this standard architecture. The peripheral devices are

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connected externally to system unit of the computer through appropriate interface and ports.

2.0 Intended Learning Outcomes

By the end of this Unit, you will be able to:

- Understand the input devices and the functions in the computer system
- Identify various kinds of input devices



3.0 Main Content

3.1 Description of Input Devices

Input devices are parts of the computer system that enable the users to enter data and the required control parameters into the computer. The input data is thereafter converted and translated into a computer-readable format for subsequent processing by the central processing unit.

Input devices can be categorized into pointing, scanning, wireless, and cable devices. Pointing input devices are used to move a cursor on the screen, examples of these include the mouse, trackball, and touchpad. The category of scanning input devices are Optical Mark Recognition (OMR), scanners, Optical Character Recognition (OCR) devices, bar code readers among others. These pointing input devices can sense and read characters directly on papers, and other kinds of materials. Bar code readers are often used to scan bar code items in the supermarkets and different kinds of industrial products in the supply chain line.

Some of the input devices include the keyboard, mouse, touchpad, webcams, microphones, joysticks, image scanners among several others. Figure 27 shows Computer with Peripheral Devices.

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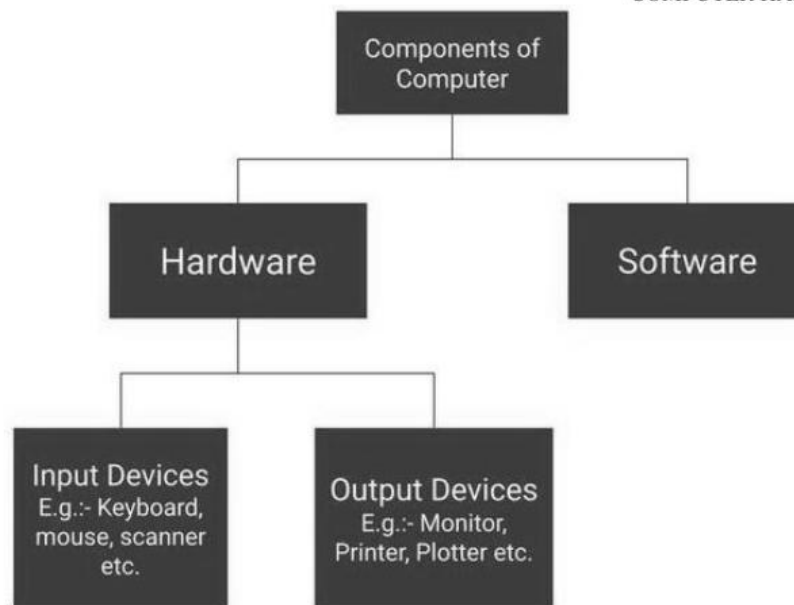


Figure 27 Computer with Peripheral Devices

3.2 Keyboard

The keyboard is the major and commonly used data entry device for different kinds of computers. Like the old typewriter machines, the keyboard is used to enter data into the computer by typing the necessary alphanumeric keys. There is a total of 104 numeric, alphabet, and function keys. The keyboard can be connected to the system unit via cable or wirelessly using Bluetooth.

Essentially, the keyboard typing keys include the following classes

- Upper and lower case alphabet keys from A-Z and a-z respectively
- Numeric keys from 0 to 9
- Special 12 function keys for specific purposes
- control (Ctrl) keys
- cursor and screen control keys are Arrow keys, page up, page down
- Alternate (Alt), Escape (Esc), Home, End, Insert, Delete
- Enter, backspace, num-lock, caps lock, shift, space bar, tab, scroll lock, sleep, print screen

Figures 28 and 30 show the illustration of computer keyboard.

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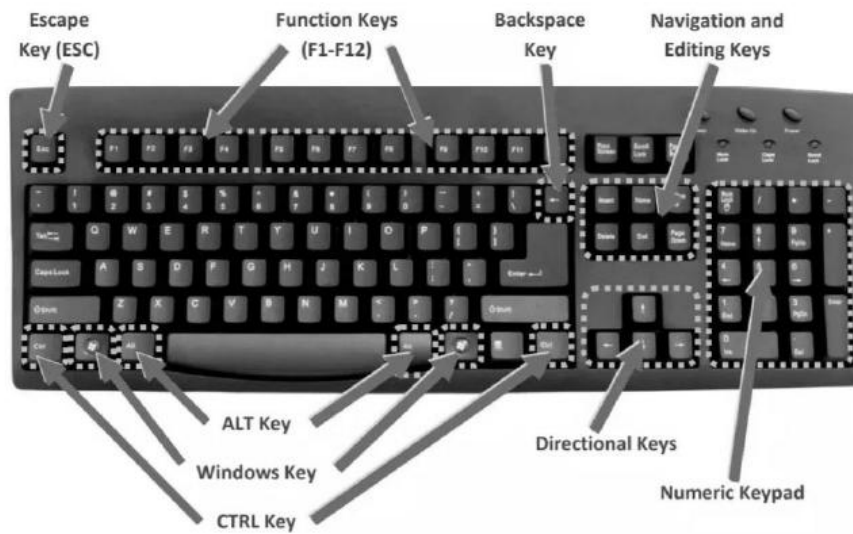


Figure 28 Description of computer keyboard



Figure 29 Computer keyboard

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3.3 Mouse

A mouse is an input pointing device that is rolled over a smooth surface to control the cursor on the computer screen through its speed controlled sensor, operational clickable buttons on the left, middle, and right parts. The functions of the buttons can be customized to select and, open files or objects. Figures 30 and 31 show the wired and wireless mouse.



Figure 31 Mouse



Figure 30 Wireless Mouse

3.4 Image scanners

The image scanner is an input device that optically scans printed materials, objects, images, and documents and thereafter converts and store them in a digital format. The scanned document be edited for further application. Figure 32 shows a scanner.



Figure 32 Scanner

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3.5 Track Ball

A trackball is a pointing device that holds a small ball held in its sensor-based socket. The sensor detects the rotational movement of the ball about its two axes. The device operates exactly like an upside-down mouse but uses space to rotate its ball. As the trackball remains stationary and the user moves the ball in various directions, it reflects activity on the screen as shown in Figure 33.



Figure 33 Track ball

3.6 Light Pen

Light pen is a light-sensitive input device that can be used to point and write or draw directly on the visual display unit, monitor, and screen. Light pen to detect raster on the screen as it passes on it with high precision. Light pen functions well like the touchscreen devices as shown in figures 34 and 35.



Figure 35 Light pen



Figure 34 Light pen on IPAD

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3.7 Headsets and Microphone

Headsets or microphones are a kind of computer voice input devices that are able to record sounds by transmitting and converting human speech or voice into electrical signals. This electrical signal is processed by the computer and the word is recognized. Application areas include in customers' service, religious programmes and technical support centers, where the users can communicate with clients while entering or typing into a computer. Figure 36 shows a typical computer microphone.



Figure 36 Computer microphone

3.8 Optical Character Reader

Optical character recognition or optical character reader (OCR) uses a low frequency light source to detect and convert drawn images, alphanumeric characters, and printed text into machine-encoded text, pictures or scanned documents. The OCR, which operates like a scanner, absorbs light by the dark areas and is reflected by the light areas. Subsequently, it reflects and receives the light by the photocells as shown in figure 37.



Figure 37 Barcode Reader

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3.9 Bar Code Reader

A barcode reader is an optical scanning device which is connected to a computer to read printed barcodes and decode the hidden data. The barcode reader which functions like a flatbed scanner, comprises of a light source, a lens and a light sensor translating for optical impulses into electrical signals. Most barcode readers contain decoder circuitry that can analyze the image data in the barcode via the sensor and thereafter send the contents of the barcode to the scanner output port.

Figure 38 and 39 show a barcode reader reading data that is coded into white and black lines.



Figure 38 Bar code reader



Figure 39 Bar code scanner

4.0 Conclusion

Several computer data input devices have been discussed. The input devices are part of peripheral devices that help users to enter data into the computer system via diverse kinds of port interfaces. Upon reception, data will be processed by the CPU for further action. Some common input devices are

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keyboard, mouse, scanner, optical character reader, barcode reader among others.

5.0 Summary

In this Unit, you have learned about some common data input devices.

6.0 Self-Assessment Exercises

What do you understand by peripheral devices?

List five types of data input devices

Discuss any 4 kinds of data input devices

Distinguish between scanner and barcode reader

7.0 References and Further Reading

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Unit 2: Output Devices

Content



1.0 Introduction

You have learned about input devices in Unit 1. In this Unit 2, you will learn more about computer peripheral devices, essentially, the output devices. These are the devices that are used to display, in human-readable form, the output of any task carried in the computer.

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2.0 Intended Learning Outcomes

By the end of this Unit, you will be able to:

- Identify some output devices often used to enter data into computers
- Differentiate between input and output devices



3.0 Main Content

3.1 Description of Output Devices

An output device is any peripheral hardware that is connected to a computer either wired or wireless so as to display, project, or physically reproduce the results of data processed by a computer. It converts electronic information into a format, which can be understood by humans.

What differentiates an input from output device is that the input device transfers data to the computer, whereas the output device receives data from the computer. Input device usage is mandatory to operate the computer, on the other hand, output devices may be optional.

Conventionally, the output devices can be grouped into data, print, visual, and sound hardware. Thus, various kinds of output hardware include visual display unit (monitor or screen), printer, plotter, headphones, computer speakers, projector, GPS, sound card, video card, braille reader, speech generating machine, among others.

3.2 Visual Display Unit

The visual display unit (VDU) popularly known as a monitor or display screen, is the main and well-known output device that displays output contents as picture elements (pixel) on the computer screen. These output contents include text, video, images, etc. The VDU looks much like a TV screen and they share similar resolution parameters.

The resolution of the monitor is determined by the number of pixels. Basically, the monitor resolution, smoothness or picture clarity, increases with more pixels. The two types of monitor screens are cathode ray tube monitors (CRT) and flat-panel screen monitors.

Cathode Ray Tube monitors, which were manufactured using old CRT technology, applied phosphorescent dots to generate pixels that form displayed images. CRT monitor screens, which were the same as the old TV CRT technology, were usually large and consumed more power. Figures 40 and 41 show LCD and CRT visual display units respectively.

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Figure 41 CRT visual display unit



Figure 40 LCD visual display unit

Flat Panel Screen Monitors

Flat-panel screen monitors, which implement a thin panel design instead of CRT technology, are much lighter, thinner, and portable. They use liquid crystals or plasma technology to generate output as light passes through liquid crystals to form the required pixels.

The three types of Flat Panel screen monitors are Liquid Crystal Devices (LCD) monitors or Non-emissive displays, Light Emitting Diode (LED) monitors or Emissive displays, and Plasma monitors. Let us explain these three types of monitor screens.

- a) Liquid Crystal Devices (LCD) monitors or Non-emissive displays
This type of flat panel display implements the light-modulating properties of liquid crystals. This technology is also replicated in LCD televisions, aircraft cockpit displays, ship cocktail screen, etc. An LCD monitor is more energy-efficient, and users' friendly. It is safer to dispose than the CRT monitors.
- b) Light Emitting Diode (LED) monitors or Emissive displays
Although the technologies of LCD and LED are similar. The LED technology of LED is more advanced than the LCD monitor. The LED monitors are lighter, thinner, and less expensive. These monitors are more reliable as they have a more broad dimming range in terms of backlighting.
- c) Plasma monitors
The picture elements in the plasma screen monitors are illuminated by a tiny bit of charged gas or plasma which is similar to a tiny neon light. These monitors are thinner, brighter, and better in performance than the Cathode ray tube, and liquid crystal display monitors.

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3.3 Printing Devices

The printing devices are specialized computerized machines that can print information permanently on materials of different forms. They are usually stand-alone systems that can be connected to computers via cable, wirelessly, or cloud. Examples of printing devices include printers, plotters, 3D printers among others. This section discusses on some printing devices.

3.3.1 Printers

A printer is the most second common type of output device that permanently transfers the processed data from the computer into a printed format and thereby produces text or images hardcopy output on paper. Today, different kinds of printers can print texts, photos and graphics objects in coloured, monochrome, and black and white. Printers used in homes and offices have high dots per inch (DPI), which generates high-quality images.

The printers are classified into Impact, and Non-Impact printers.

a) Impact printer

This is old type of printer noisily prints characters by striking on the ribbon, and thereby imprint on the paper.

The impact printers can either be character or line printers.

i. Character printers

These types of printers are capable of printing text, and a single character at a time in the average speed is up to 300 lines per minute. The different types of character printers include chain, band, dot matrix, and daisy wheel printers. The commonly used printers are dot matrix and daisy wheel.

ii. Line Printers

Line printers can print and display output texts line by line. The two types of line printers are drum printers and chain printers. Basically, the drum printer has high speed and can print up to 300 to 2000 lines per minute.

b) Non-Impact printer

The non-impact printers can print without striking the ribbon. These include laser printers and inkjet printers.

i. Laser Printers

Laser printers which use a photoelectric drum, powdered ink, and laser light to produce dots to form the characters for printing. When a print command is given, a laser beam draws the document on a selenium-coated drum using an electrical charge, giving a clean copy of the image on the paper. They have fewer smearing

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problems than an inkjet printer since ink is not used. A laser printer uses the same technology and procedures like a copying machine. Figures 42 and 43 show LaserJet printers.

ii. Inkjet Printers

This printer generates output by spraying droplets of ink onto the paper through a small nozzle. Inkjet printers are the most widely used type of printer. There are expensive and inexpensive models available in the market. It was the most common type of printer in the olden days due to limited available technology. The printers are less expensive.



Figure 42 LaserJet Printer



Figure 43 Coloured Epson Printer

3.4 Graph Plotter

A plotter output device, which is similar to a printer, but usually bigger in size, can be used to draw large high quality, images, pictures and vector graphics.

A plotter is used to generate and draw hardcopy on papers large drawings, architectural blueprints, engineering drawings, graphic cards, cadastral maps, satellite imageries among others. Figure 44 shows a graph plotter.

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Figure 44 Graph Plotter

3.5 Computer Speakers

Computer speakers are one of the most common output hardware used with a computer. The speakers receive audio as input from the computer via a sound card. The internal amplifiers of the speaker can be used to control or vary the volume or the amplitude of the sound. Connection of external speakers can be used to enhance the volume and other in-built parameters of the sounds. Figure 45 shows computer speakers.



Figure 45 Computer speakers

4.0 Conclusion

Computer output hardware are quite important in showing and revealing the results of processing by the CPU. Some common output devices are visual display units, printers and output accessories. A printer is a peripheral output device which produces a hardcopy of graphics or text from a computer usually on papers. While most output is human-readable, bar code printers are an example of an expanded

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use for printers. Different types of printers include 3D printers, inkjet printers, laser printers, and thermal printers.

5.0 Summary

In this Unit, you have learnt about some common computer output devices. Their usage enables the users to view and appreciate the results of processing.

6.0 Self-Assessment Exercises

- Differentiate between CRT and flat panel screen monitors
- Mention the three types of Flat Panel screen monitors

7.0 References and Further Reading

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<https://www.classmate4u.com/output-devices-of-computer/>

Module 4: Computer Hardware Components**Unit 1: System board - Motherboard**

Content



1.0 Introduction

In this Unit, you will learn about the system motherboard. The motherboard is the main circuit board firmly fixed inside a computer system unit to which all other internal electronic components and interfaces are connected via its ports. Among these are CPU, RAM slots, controllers, system chipset, ROMs, communication cables.



2.0 Intended Learning Outcomes

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By the end of this Unit, you will be able to:

- Understand details about computer hardware components
- Get familiar with the design of computer motherboards



3.0 Main Content

3.1 Description of Motherboard

The motherboard is a major large integrated circuit component in the system unit. It is rectangular in shape and contains chipsets, switching devices, onboard ports, sockets, slots, ROM chips, and interfaces for interconnecting CPU, hard drives and optical drives, CMOS, data buses, cooling fan, power supply unit, and secondary storage devices. Different kinds of communication and power cables are used for appropriate connections. Expand cards and other peripheral card slots also provide connections to video, speakers, and sound interfaces. The cooling fans and heat sink prevent excessive heat radiation and thereby maintain the specified operating condition inside and outside the system unit.

Some popular manufacturers of the motherboard include Intel, ASUS, AOpen, ABIT, Biostar, Gigabyte, MSI. Figures 46 and 47 show computer motherboard.

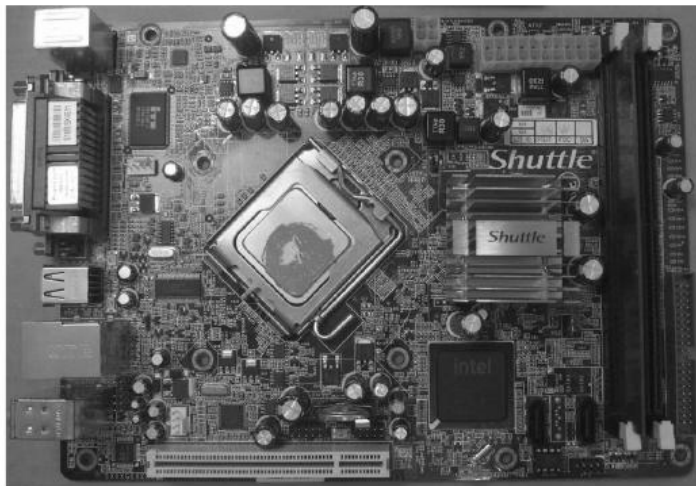


Figure 46 Computer motherboard

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The chipset is one of the most important components in the motherboard. Older motherboards were designed with a lot of different chips scattered all the motherboard. There are chips for different things like chips for bus controller, memory controllers, keyboard controllers, etc. Thus, diver chips control different functions on the motherboard. A chipset is a smaller set of chips that has replaced a larger amount of chips so as to control data flow between the CPU, the peripherals, bus slots, memory, and other parts of computer.

Furthermore, the advancement in technology enable the chips manufacturers to reduce the number of chips and rather centralizes somewhere to execute same job and localize them to chipset. Buses connect the CPU to various internal components and to expand cards for graphics and sound. The CMOS battery attached to the motherboard provides the memory power backup for system clock, date, time and other system setup parameters in the BIOS chip. The CMOS battery is mostly CR2032.

It is important that video cards, hard disks, sound cards, power supply modules, and system unit cases are compatible and interoperable with the motherboard in order for the entire computer system to function properly as required.



Figure 47 Motherboard components layout

The computer system clock is an electrical pulse generator that sends out a pulse of electricity at regular intervals. The electronic components of the computer require these electric pulses in order to execute tasks. The more pulses sent out by the system clock in a cycle, the faster the CPU works. The first personal

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